

**Homework #1**  
**EECS 41104/51104 Embedded Systems**  
**Due August 28<sup>th</sup>, 2026**

**Problem #1.**

1. Sign in or create an account at [learn.zybooks.com](https://learn.zybooks.com)
2. Enter zyBook code: UARKCSCE41104AndrewsFall2026
3. Subscribe

This will be graded by viewing if you have answered participation activity 1.1.3 on Moore's Law.

**Problem #2.**

List three characteristics of embedded systems not found in general purpose systems.

**Problem #3.**

Describe the major IP components of the baseline System on Chip (SoC) architecture that you will create in the laboratory.

**Problem #4.**

Consider a chip in a smartphone today. If Moore's Law holds, that chip in 6 years would hold how many times more transistors? (hint: from the book)

**Problem #5.**

Consider a chip in a smartphone today. If Moore's Law holds, the transistors within that chip in 12 years would occupy what fraction of the chip's current size? Write answer as a fraction in the form: 1 / 4 (hint: from the book)